

Task 1: Natural Language Processing (NER)

Currently, the custom model focuses only on extracting the mountain's name. Since pre-trained NLP models generally already recognize standard location tags (like LOC or GPE), the next step is to improve the dataset to capture the relationship between the mountain and its surrounding region. Since different mountains can share the exact same name, extracting this combined context is crucial to uniquely identify the peak. Ultimately, linking the mountain name with its regional context would allow the system to perform **entity linking** – automatically matching the text against a **knowledge base** (like Wikipedia) to find the exact real-world object.

Task 2: Computer Vision (Sentinel-2 image matching)

Currently, splitting large satellite images into smaller tiles is done sequentially. To significantly speed up this data preparation step, I would implement parallel processing. By utilizing multiple processes (multiprocessing), the system can use all available CPU cores to cut multiple tiles simultaneously.